



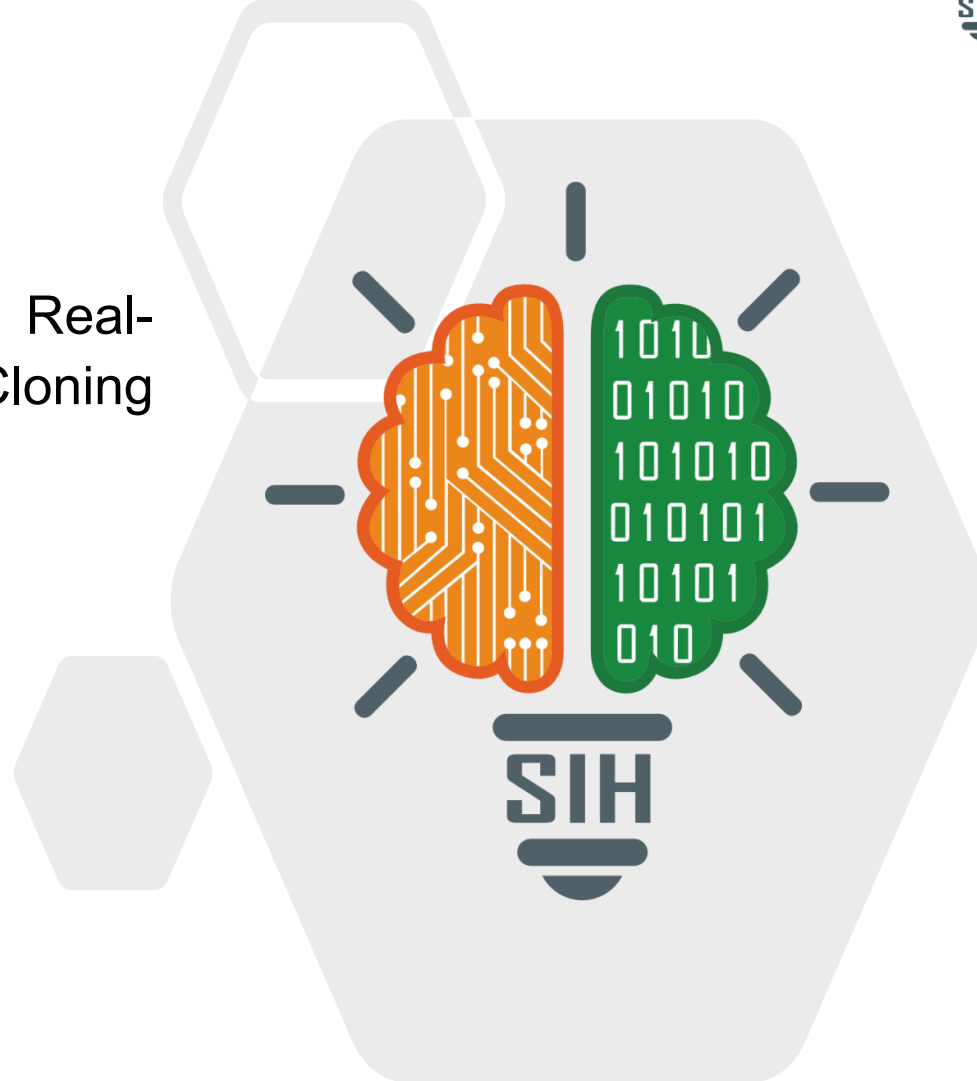
VICTRIX

# SMART INDIA HACKATHON 2026



SMART INDIA  
HACKATHON  
2026

- **Problem Statement ID** – SIH26104
- **Problem Statement Title-** AI-Powered Real-Time Detection and Prevention of Voice Cloning Impersonation Attacks
- **Theme-** Blockchain & Cybersecurity
- **PS Category-** Software
- **Team Name** - Victrix



## PROBLEM OVERVIEW

*Why existing checks fail*

- Voice cloning from a few seconds of audio now enables convincing CXO / official impersonation.
- Caller ID and manual call-back checks can no longer separate real callers from AI-generated voices.
- Attacks are increasingly delivered over VoIP, mobile, and enterprise collaboration platforms.
- No real-time mechanism exists today to flag a cloned voice while the call is still happening.
- High-pressure social-engineering calls push victims into approving transfers before they can verify.

## PROPOSED SOLUTION

*Idea / Solution / Prototype*

*A real-time voice integrity framework that analyzes live call audio with deep learning + signal processing to detect AI-cloned voices.*

- **Real-Time Risk Scoring** – analyzes live call audio and computes a continuously updated impersonation risk score.
- **Multi-Layer Voice Forensics** – acoustic, spectral, and prosody analysis detects synthesis artifacts neural TTS leaves behind.
- **Instant Alerting** – flags high-risk calls before fund transfers or confidential disclosures are approved.
- **Privacy-Preserving** – on-device / edge inference minimizes central storage of raw voice data.
- **Built for India** – language-agnostic features with acoustic models tuned for Indian languages and accents.

## INNOVATION & UNIQUENESS

Real-time streaming detection (not offline analysis) with an explainable, continuously evolving risk score — built for Indian multilingual / accent diversity and privacy-preserving on-device inference.

## METHODOLOGY

1

### Data & Benchmarking

Train/evaluate on ASVspoof, WaveFake & Fake-or-Real datasets; establish EER baseline.

2

### Model Development

Build spoof-detection classifier (RawNet2 / AASIST / Wav2Vec2 front end).

3

### Streaming Pipeline

Chunked audio ingestion via WebRTC/WebSocket for near real-time inference.

4

### Risk Scoring Engine

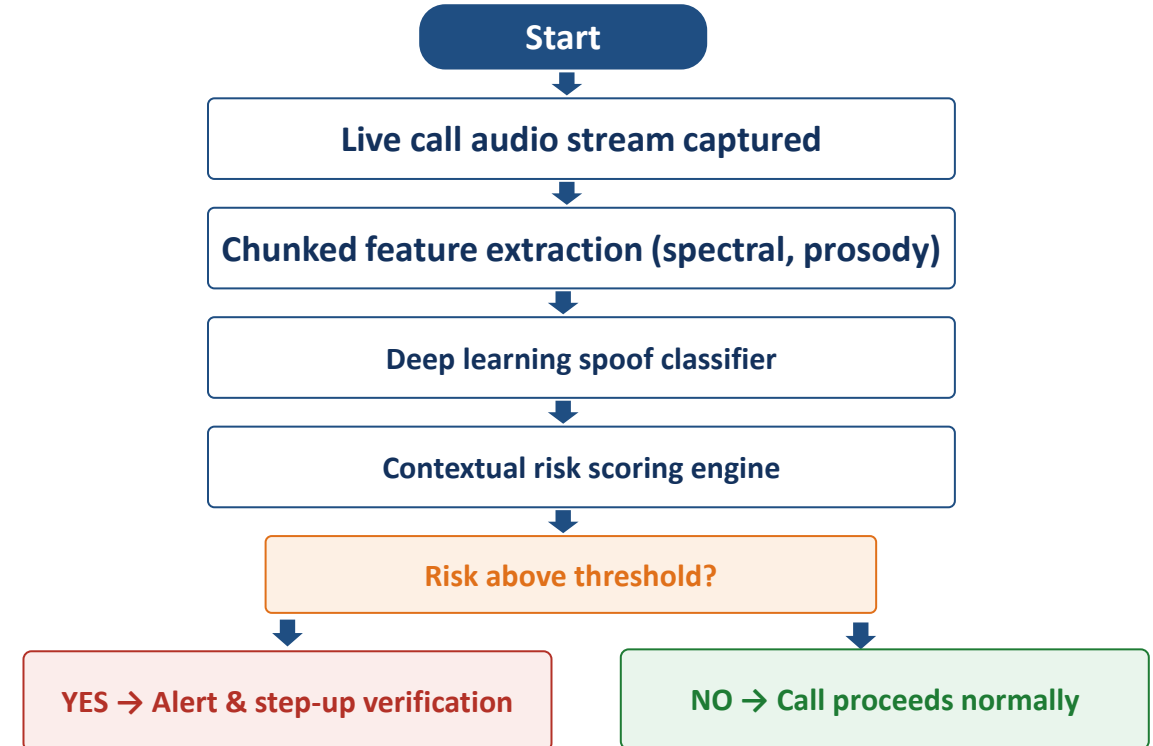
Blend model output with call metadata into one evolving risk score.

5

### Alerting Layer

Threshold-based UI/SMS/webhook alerts before high-risk actions proceed.

## DETECTION FLOW



**TECH STACK:** PyTorch • torchaudio • Wav2Vec2 / RawNet2 / AASIST • Librosa + SciPy • FastAPI / Node.js • WebRTC / WebSocket • React dashboard • Docker • ONNX Runtime (edge)



# FEASIBILITY AND VIABILITY



## ANALYSIS OF FEASIBILITY

### 1. TECHNICAL

Public benchmarks (ASVspoof, WaveFake, FoR) and published architectures (RawNet2, AASIST) make a real, evaluable model achievable.

### 2. DATA

Labeled spoof/bonafide datasets are freely available – no synthetic or self-generated ground truth needed.

### 3. DEMO

A live audio play-and-flag demo is achievable within the hackathon window with a lightweight model.

### 4. INTEGRATION

REST/gRPC APIs make the detection core pluggable into banking, telecom, or enterprise call flows.

### 5. PRIVACY

On-device / edge inference options align with data-protection expectations for voice biometric data.

### 6. SCALABILITY

Streaming architecture (chunked inference) generalizes from a single demo call to production call volumes.

## POTENTIAL CHALLENGES & RISKS

### True Real-Time Latency

Running detection with negligible delay on live streaming audio is a genuine engineering challenge.

### Cross-Accent Generalization

Detector accuracy may vary across India's diverse languages and accents.

### Scope Creep

Banking + telecom + enterprise integration is too broad to build fully in the hackathon window.

## STRATEGIES FOR OVERCOMING

### Lightweight Live Model

Short overlapping audio windows + a lightweight model tuned specifically for the live demo.

### Scoped Claims

Limit accent/language claims to the data actually used in training – no overclaiming.

### One Reference Integration

Build a single simulated banking call-approval flow instead of all three verticals.



# IMPACT AND BENEFITS



*Protects banks, enterprises, government agencies, and individuals from voice-cloning-driven financial fraud and social engineering attacks.*

## IMPACTS



### Fraud Prevention

Stops voice-cloning-driven CEO fraud and vishing before funds move.



### Restored Trust

Rebuilds confidence in voice-based communication for banks and enterprises.



### Early Warning

Flags impersonation attempts during the call, not after the damage is done.



### National Resilience

A reusable detection layer strengthens India's telecom/enterprise cyber defenses.

## BENEFITS



### Economic

Reduces direct financial losses from CEO-fraud and vishing scams.



### Social

Restores public trust in voice-based communication channels.



### Security

Reusable detection layer for telecom and enterprise cyber resilience.



### Deployability

API-first design plugs into existing call infrastructure without rebuilds.

1

**ASVspoof Challenge Series (2019 / 2021 / 2024)**

Standard, peer-recognized benchmark for spoof/anti-spoofing detection with labeled bonafide vs. spoofed audio.

Link: [asvspoof.org](https://asvspoof.org)

2

**Jung et al., “AASIST: Audio Anti-Spoofing Using Integrated Spectro-Temporal Graph Attention Networks” (ICASSP 2022)**

State-of-the-art end-to-end anti-spoofing architecture – core reference for the detection model.

Link: [arxiv.org/abs/2110.01200](https://arxiv.org/abs/2110.01200)

3

**Tak et al., “RawNet2-based Anti-Spoofing for Automatic Speaker Verification” (ASVspoof 2021 workshop)**

Raw-waveform end-to-end spoof detector – alternative/ensemble architecture reference.

Link: [asvspoof.org/asvspoof2021](https://asvspoof.org/asvspoof2021)

4

**WaveFake & Fake-or-Real (FoR) Datasets**

Supplementary datasets covering diverse TTS/vocoder generation methods for broader model robustness.

Link: [github.com/RUB-SysSec/WaveFake](https://github.com/RUB-SysSec/WaveFake)